

A stem and a leaf

A stem-and-leaf graph, or stemplot, splits each value into a stem and a final significant digit called a leaf.

For the value 47, what stem and leaf could we use?

Exploratory data analysis

Exploratory data analysis uses graphs and summaries to look for patterns and unusual observations.

What would you look for first in a plot of download durations?

Reading the key

Synthetic file sizes. Key: 3 | 4 means 34 KB.

Stem	Leaves
2	1 5 5
3	0 4 8
4	2

What values does this stemplot display?

Build a stemplot

Synthetic file sizes (KB): 21, 25, 25, 30, 34, 38, 42.

How would you organize these values into a stemplot with a key?

Repeated values

Synthetic file sizes. Key: 2 | 5 means 25 KB.

Stem	Leaves
2	1 5 5
3	0 4 8
4	2

How many files have size 25 KB?

Decimal leaves

Synthetic response times: 1.2, 1.5, 1.5, 2.1, 2.8 seconds.

What key and stemplot would preserve every recorded value?

An empty stem

Synthetic file sizes (KB): 12, 15, 17, 31, 35.

Why should a stemplot include the stem 2?

A possible outlier

An outlier, or extreme value, does not fit the overall pattern. Synthetic times: 1.1, 1.2, 1.3, 1.4, 1.6, 8.9 seconds.

Which value deserves investigation?

Side-by-side stemplots

A side-by-side (back-to-back) stemplot shares stems between two sets of leaves.

Why should both groups use the same stem scale?

Keeping an unusual value

A recorded file size is much larger than all others.

What would you check before deciding whether to remove it?

Comparing two groups

Synthetic sizes (KB):

A: 12, 14, 16, 21, 23

B: 18, 21, 24, 27, 29

How could a back-to-back stemplot compare the two groups?

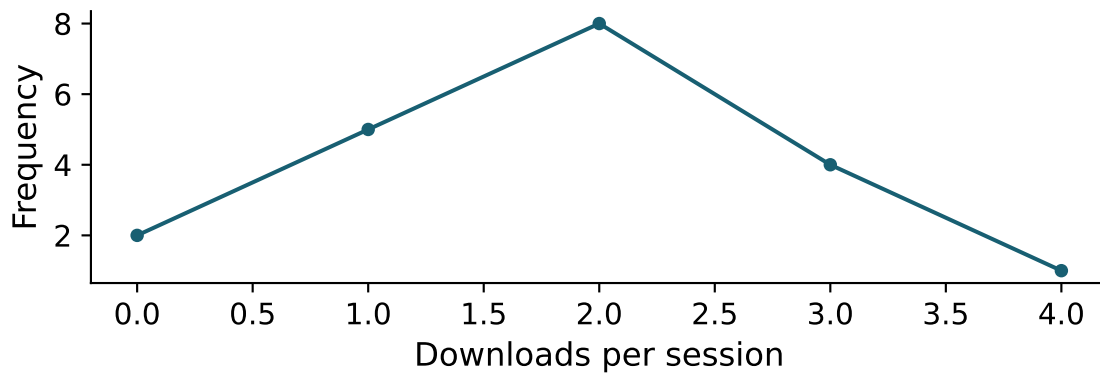
Frequency line graphs

A frequency line graph connects frequency points above numerical data values.

How does its horizontal axis differ from a bar graph's categories?

A line graph

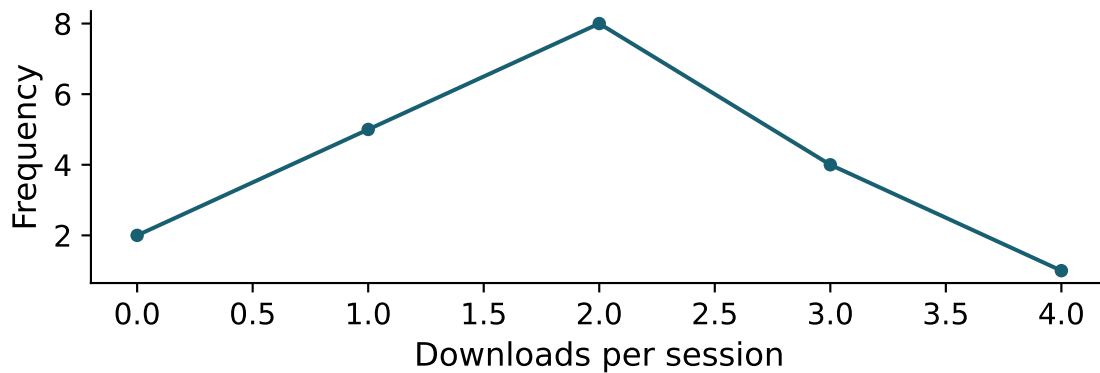
Synthetic downloads per session.



What does the peak show?

Frequency from a graph

Synthetic downloads per session.



How many sessions had at least three downloads?

Build a frequency line graph

Synthetic downloads per session.

Downloads	Sessions
0	2
1	5
2	8
3	4
4	1

Where should each frequency point be plotted?

A bar graph

A bar graph uses separate bars for categories. Synthetic notebook counts: blue 12, black 8, red 5.

What should the axes show?

Counts or proportions?

Synthetic notebook counts: blue 12, black 8, red 5.

What height should the blue bar have on a relative-frequency graph?

Category order

Synthetic notebook counts: blue 12, black 8, red 5.

Would changing the category order change the data being displayed?

Constructing a bar graph

Synthetic notebook proportions: blue 48%, black 32%, red 20%.

How would you draw a labeled bar graph with a percentage axis?

Choose a display

A spreadsheet contains printer brands and file sizes in KB.

Which variable is suited to a stemplot?